2027 Guide to AI in Science

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Table of contents

Preface

AI is radically redefining the nature of scientific inquiry — for better and for worse.

In Fall 2026, faculty, graduate students, and postdocs across multiple departments at the University of Florida came together to understand this urgent development. In a seminar titled *Collective [Artificial] Intelligence in Science*, we documented and discussed how AI is changing each component of our research workflow. This includes project management, idea development, data analysis, writing, and dissemination. To push our understanding, we alternated between the perspective of an AI zealot and an AI critic – identifying key resources, consensus statements, and areas of disagreement along the way. The result is the **2027 Guide to AI in Science**.

The seminar began with an introduction to big-team science – its rise[1], competitive advantages[2], and relevance to the current moment. For instance, the paper describing OpenAI's GPT-1 had 4 authors; the paper describing GPT-3 had 31 authors; and the paper describing GPT-5 had 483 authors (Figure 1). From there, the course itself became a large-scale collaboration, dedicated to understanding how these AI advancements are shaping science.

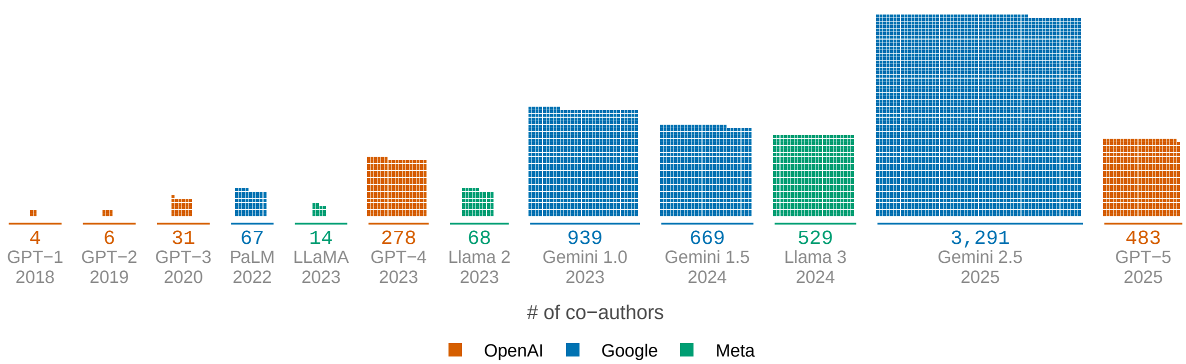

Figure 1: Credited co-authors on flagship AI model reports from OpenAI, Google, and Meta, 2018–2025. One square is one credited person; each block's area is its author list.

Each week followed a consistent structure (Figure 2). First, we would select a component of the research pipeline to examine (e.g., data analysis). Next, we would spend one week compiling resources for understanding the emerging role of AI in that part of the research pipeline. In class, we engaged in peer-to-peer teaching – all with the goal of identifying the top 3 resources to include in the guide. Last, we used a live onlined voting platform to solicit, debate, and vote upon key takeaways for the week (https://pol.is/). The result is the **2027 Guide to AI in Science**: a summary of our top resources, takeaways, and disagreements.

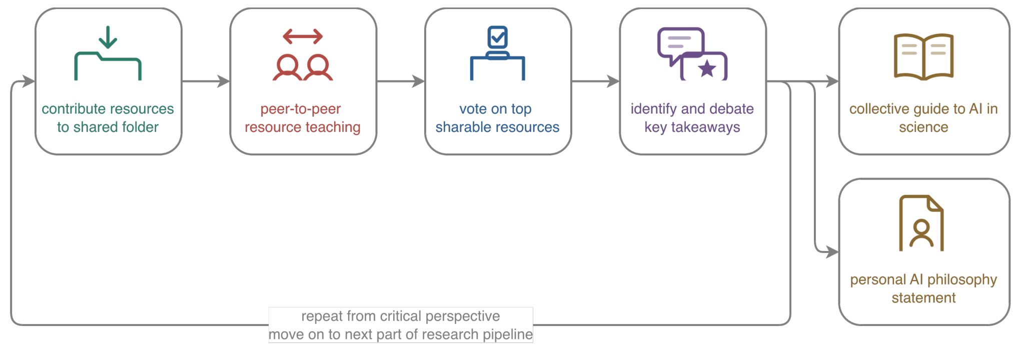

Figure 2: Structure of the University of Florida Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science.

Project Management

Project Management: Using AI

> *How is AI changing the way research teams plan, coordinate, and keep track of their work?*

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

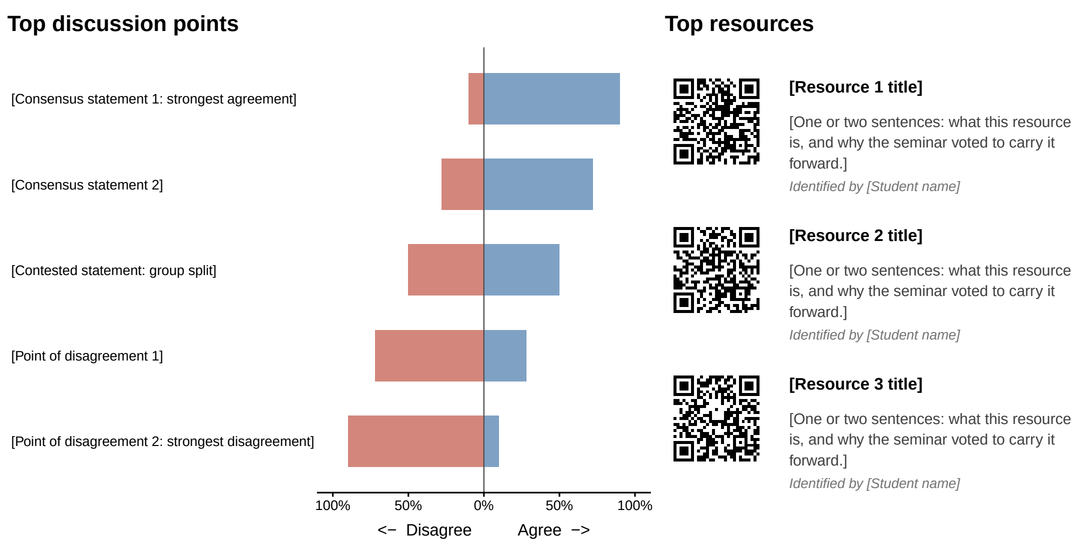

Figure 1: What the seminar concluded about using AI for project management. Left: statements put to the group, ordered from strongest agreement to strongest disagreement. Right: the three resources voted forward, each linked by QR code and credited to the student who surfaced it.

> **Wireframe.** Figure 1 is drawn from placeholder rows in `week-summary/data/week-01-*.csv`. The bars, titles, and names are layout stand-ins, not results. Replace the two CSVs and the figure fills itself in.

The Case

[Open with one concrete situation — a real project, told in a paragraph or two. Something a reader in this seminar would recognize: a multi-site study whose coordination overhead outgrew the science, a lab notebook nobody kept current, a grant timeline that slipped. Establish what the coordination problem actually cost before naming any tool.]

[Then the turn into the week: what an AI-assisted version of this situation would look like, and which of the questions below it raises. This paragraph is the transition — it should leave the reader holding a specific question that the resources and the votes then answer.]

Top Resources

The three resources the seminar voted to carry forward, in the order they appear in Figure 1.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation] · **Identified by [student name]**

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year] · **Identified by [student name]**

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year] · **Identified by [student name]**

What it is:

Why it made the cut:

Who it is for:

Where We Agreed

The statements that reached consensus in the Polis session — the upper bars of Figure 1.

[One short paragraph framing what the agreement has in common. If the consensus statements share a shape — say, agreement on disclosure but not on delegation — name it here rather than leaving the reader to infer it.]

1. *[consensus statement — % agreeing]*
2. *[consensus statement — % agreeing]*

Where We Did Not

The statements that split the group — the lower bars of Figure 1.

[One short paragraph on the nature of the disagreement. Whether it is empirical (we disagree about what is true) or normative (we disagree about what is acceptable) is the useful distinction, because only the first kind is settled by evidence.]

- *[position vs. counter-position — and what evidence would move it]*
- *[position vs. counter-position — and what evidence would move it]*

Back to The Case

[Return to the opening situation and re-run it against what the week established. What would the group's consensus have changed about how that project was managed? What would it not have changed? Where the group disagreed, say plainly that the case does not settle it either.]

[Close on the handoff to Chapter — this chapter took the receptive lens; the next one takes the critical one on the same stage of the pipeline.]

Open Questions

- *[question the week raised but could not answer]*

Project Management: Critiquing AI

What breaks when the coordination of a research team is delegated to AI?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review

Literature Review: Using AI

How is AI changing the way we find, triage, and synthesize prior work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review: Critiquing AI

What does AI-mediated search hide, distort, or invent — and how would we notice?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Idea Development

Idea Development: Using AI

How is AI changing the way research questions and hypotheses are generated?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Idea Development: Critiquing AI

What happens to originality and intellectual ownership when ideas are co-generated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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[Author(s), year]

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The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Data Collection

Data Collection: Using AI

> *How is AI changing the way we recruit participants, build instruments, and gather data?*

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Data Collection: Critiquing AI

What threats to data integrity, validity, and research ethics does AI introduce?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis

Analysis: Using AI

How is AI changing the way we write code, model data, and interpret results?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

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2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis: Critiquing AI

Which analytic errors survive when the analysis is delegated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing

Writing: Using AI

How is AI changing the drafting, revising, and labor of scientific writing?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing: Critiquing AI

What is lost when writing stops being thinking — and who counts as an author?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination

Dissemination: Using AI

How is AI changing peer review, publishing, and how findings reach their audience?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination: Critiquing AI

What does AI do to trust, gatekeeping, and the integrity of the scientific record?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Personal AI Philosophy Statements

Having examined the research pipeline together, where does each of us now stand?

The preceding chapters record where the seminar converged as a group. This chapter records where its members stand individually.

Alongside the shared guide, participants wrote personal AI philosophy statements: short, signed accounts of how each of us intends to use — and refuse to use — AI in our own research. Each statement is reproduced in full below and is free to share, quote, or adapt.

The statements

Nicholas A. Coles

[Role, department, institution]

[Statement.]

[Author]

[Role, department, institution]

[Statement.]

Where we agree

Commitments that recur across the statements.

1. *[point of agreement]*
2. *[point of agreement]*
3. *[point of agreement]*

Where we disagree

The substantive splits — not differences of emphasis, but positions that cannot both be right.

- *[position vs. counter-position — and what would settle it]*

Each statement speaks only for its author. Disagreement between them is the point, not an error to be corrected.

References

1. Coles, N. A., Hamlin, J. K., Sullivan, L. L., Parker, T. H. & Altschul, D. Build up big-team science. *Nature* **601**, 505–507 (2022).
2. Coles, N. A., Goes Braga Takayanagi, J. F., Grant, G. L. & Basnight-Brown, D. M. The rise, impact, and imbalances of big-team psychology. *Psychological Science* **37**, 287–297 (2026).